LECTURE 2 · INSTRUCTOR SHEET

Autoregression on four pixels

A compact worked example for the blackboard. The point is the chain rule: a fixed image has an exactly evaluable probability, while generating an image remains stochastic and sequential.

1 · Draw the observed image and the scan order

x1 = 1	x2 = 0	x3 = 1	x4 = 1
first	condition on x1	condition on x1,x2	condition on x1,x2,x3

$$p(x1,x2,x3,x4) = p(x1) \cdot p(x2 \mid x1) \cdot p(x3 \mid x1,x2) \cdot p(x4 \mid x1,x2,x3)$$

2 · Use the model's four conditional probabilities

Observed value	Model output used	Factor
x1 = 1	p(x1=1) = 0.80	0.80
x2 = 0	p(x2=0 | x1=1) = 1 - p(x2=1 | x1=1) = 1 - 0.25	0.75
x3 = 1	p(x3=1 | x1=1,x2=0) = 0.70	0.70
x4 = 1	p(x4=1 | x1=1,x2=0,x3=1) = 0.90	0.90

3 · Multiply; then take the negative log

Image likelihood	**$0.80 \times 0.75 \times 0.70 \times 0.90 = 0.378$**
Negative log-likelihood	**$-\ln(0.378) = 0.9729 \approx 0.973$ nats**

Exact likelihood	For this fixed complete image, all normalized factors can be evaluated and multiplied. No partition function, variational bound, or Monte Carlo estimate is needed.
Not deterministic generation	Sampling still draws x1, then x2 conditioned on the sampled x1, and so on. A different random seed can yield a different image. Greedy decoding would be a separate choice.

Blackboard prompts

1. Ask why the second factor is 0.75 rather than 0.25. 2. Ask which later conditionals may change if x2 changes. 3. Contrast parallel likelihood evaluation during training with sequential sampling at inference. 4. End with: exact probability of a fixed image does not mean the generated image is fixed.